Spectral Theorem

Given A self-adjoint, we have 2 related goals:

1) diagonalization: define a measure space X, μ and $Q: L^2(X, d\mu) \to \mathcal{H}$ so that $A = Q M_\alpha Q^{-1}$ for some $\alpha: X \to \mathbb{R}$

2) functional calculus: given a bounded Borel fn. f on $\mathbb{R}$ define $f(A) \in \mathcal{L}$

Strategy: start with unitary op's (related to self-adjoint by the Cayley transformation)

1) Use Fourier series to define continuous unitary functional calculus
2) Use unitary functional calc. to build a measure
3) Transfer this to the self-adj. case

Unitary operators

A unitary map is a bijective isometry, and for Hilbert space, isometry $\iff$ preserves inner product

$$\langle Uf, Ug \rangle = \langle f, g \rangle \quad \forall f, g$$

$\Rightarrow$ Lemma: U is unitary $\iff$ $U^* U = U U^* = 1$

Exercise: for unitary U, $\sigma(U) \subset S^1 :=$ unit circle in $\mathbb{C}$

Smooth functional calc.

Let $f \in C^\infty(S^1)$.

The Fourier series representation converges uniformly

$$f(e^{i\theta}) = \sum_{k \in \mathbb{Z}} \hat{f}(k) e^{ik\theta} \quad \text{where} \quad \hat{f}(k) = \frac{1}{2\pi} \int_0^{2\pi} e^{-ik\theta} f(e^{i\theta}) \, d\theta$$

(f smooth $\Rightarrow |\hat{f}(k)| = O(|k|^{-N})$ for all N)

For U unitary, define

$$f(U) := \sum_{k \in \mathbb{Z}} \hat{f}(k) U^k$$

Since $\|U^k\| = 1 \ \forall k$, this converges in $\mathcal{L}$

Thm (continuous unitary functional calc.)
For U unitary, the map $f \mapsto f(U)$ extends to a continuous map $C^0(S^1) \to \mathcal{L}$ satisfying

(a) $f(U)^* = \bar{f}(U)$

(b) $f(U) g(U) = fg(U)$

(c) If $f \geq 0$, then $f(U) \geq 0$

(d) $\|f(U)\| \leq \sup |f|$

Proof: First prove the properties for $f \in C^\infty$

(a) $$f(U)^* = \sum_k \overline{\hat{f}(k)} (U^k)^*$$
$$= \sum_k \overline{\hat{f}(k)} U^{-k} \quad \text{since } U^* = U^{-1}$$

Check $\widehat{\bar{f}}(k) = \hat{f}(-k)$

$\Rightarrow f(U)^* = \bar{f}(U)$

(b) $$f(U) g(U) = \sum_{k,l} \hat{f}(k) \hat{g}(l) U^{k+l}$$
$$= \sum_{k,n} \hat{f}(k) \hat{g}(n-k) U^n$$
$$= \sum_n f * \hat{g}(n) U^n$$

Check: $\widehat{fg} = \hat{f} * \hat{g}$

(c) Given $f \geq 0$, approximate $|f|$ by $h_\varepsilon := \sqrt{f + \varepsilon}$ for $\varepsilon > 0$

$h_\varepsilon \in C^\infty(S^1, \mathbb{R}) \Rightarrow h_\varepsilon(U)$ is self-adjoint by (a)

$$\langle v, f(U)v \rangle = \langle v, (h_\varepsilon(U)^2 - \varepsilon) v \rangle \quad \text{for } v \in \mathcal{H}$$

$$= \|h_\varepsilon(U)v\|^2 - \varepsilon \|v\|^2$$

$$\Rightarrow \langle v, f(U)v \rangle \geq -\varepsilon \|v\|^2 \quad \forall \varepsilon > 0$$

(d) Let $M := \sup |f|$

$$M^2 - |f|^2 \geq 0 \Rightarrow (M^2 - |f|^2)(U) \geq 0 \text{ by (c)}$$

$$\Rightarrow \langle v, (M^2 - |f|^2)(U) v \rangle \geq 0$$

$$M^2 \|v\|^2 \geq \|f(U)v\|^2 \text{ by (a) \& (b)}$$

$$\therefore \|f(U)\|_{op} \leq M$$

The extension to C^0 now follows from (d):
Given $f \in C^0$, take $f_n \in C^\infty$ s.t. $f_n \to f$ uniformly on S^1
($C^\infty(S^1)$ is dense in $C^0(S^1)$ w.r.t. sup norm, by the Stone-Weierstrass thm)

By (d), $f_n(U)$ is Cauchy in $\mathcal{I}$, so we can set

$$f(U) = \lim_{n \to \infty} f_n(U)$$

Easy to check this is indep. of (f_n) and (a), ... (d) extend to the limit.

Measure construction

Thm (Riesz representation)
Given a compact metric space X and a positive linear functional $\beta : C^0(X) \to \mathbb{C}$, there exists a unique Borel measure μ on X such that

$$\beta(f) = \int_X f \, d\mu \quad \text{for} \quad f \in C^0(X)$$

Here, positive means $f \geq 0 \Rightarrow \beta(f) \geq 0$

positive & linear $\Rightarrow$ bounded, hence continuous

Steps in the proof
1) For U open $\subset X$, define

$$\mu_0(U) := \sup \{ \beta(f) : f \in C^1(X),\ 0 \leq f \leq 1,\ \operatorname{supp} f \subset U \}$$

2) For $A \in \mathcal{B}(X)$ (Borel sets), define

$$\mu(A) := \inf \{ \mu_0(U) : U \text{ open and } A \subset U \}$$

3) Check that μ is countably additive and $\int_X f \, d\mu = \beta(f)$ for $f \in C^1$.